

Interaction Obstructions, Resonant PT Breaking, and Doubled Jordan Symmetry in Fourth-Order Theories

GPT-5.6.sol*

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Abstract

We formulate a deformation problem for the canonical positive pseudo-Hermitian metric of the Pais–Uhlenbeck oscillator and for its Krein/ghost-parity alternative, and follow both into interacting fourth-order theories. Four results organize the paper. (i) The positive metric deforms perturbatively at generic frequencies, with explicit Hermitian generators through third order. (ii) At internal rational resonances the deformation is obstructed by exact on-shell conversion operators, and at the 3:1 resonance the obstruction is spectral: an excited resonant multiplet acquires a complex-conjugate energy pair at second order — exact, by a Sturm count of the effective resonant-shell characteristic polynomial, and reproduced by direct diagonalization of the truncated Hamiltonian; the statement for the full unbounded operator remains open — so that no positive invariant metric exists where the pair persists. (iii) In the perfect-square field theory the momentum continuum turns such isolated resonances into generic scattering shells: the branch-changing process $H + L \rightarrow L + L$ obstructs the analytic second-order deformation of the pointed-sector positive metric on an open subset of its shell, with an exact value at a rational kinematic point. (iv) The exact two-field rewriting $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$ converts the theory into $-\partial U \cdot \partial V + \frac{\lambda^2}{2}U^2V^2$ and exhibits an exact exchange symmetry $U \leftrightarrow V$ of the extended two-field theory, which maps two oppositely oriented null Jordan sectors into one another and whose linearization is the bounded confluent limit of the regulated branch parity on the doubled space — a structure that survives precisely where pointed positive completions fail. All finite-dimensional and modewise identities are machine-verified; derivations of the principal results are given in the appendices.

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1 Introduction

A companion series [5, 6, 7, 8] analyzed the *free* fourth-order theories — the Pais–Uhlenbeck (PU) oscillator, the fourth-order scalar field, and linearized quadratic gravity — and established a three-level structure: one complex spectral covariance, several real forms (involutions), and several completions. The positive pseudo-Hermitian completion of Bender and Mannheim [1, 2] and the Krein/ghost-parity completion of Bateman and Turok [3] are real forms of one free theory, distinguished at Jordan degenerations. This paper begins the interacting classification: which of these free structures can be continuously deformed when nonlinear processes are switched on?

The organizing mechanism, established here first for discrete oscillator resonances and then for continuum scattering shells, is

metric deformation $\longrightarrow$ on-shell cohomological obstruction $\longrightarrow$ spectral complexification.

Results. For the PU oscillator with the cubic vertex $V = -iy^3$ inherited from a real z^3 interaction: the deformation exists at first order for all $\omega_1 > \omega_2 > 0$ and at second and third order for generic frequencies (Sections 3–4); it is obstructed at second order at $\omega_1 = 3\omega_2$ by the exact on-shell conversion class $\propto a_1 a_2^{\dagger 3} - a_1^{\dagger} a_2^3$, gauge-independently (Section 5); the obstruction is spectral — a complex-conjugate pair appears in the $E = 27\omega_2$ multiplet at $O(\zeta^2)$, certified exactly (Section 6); a third-order obstruction appears at $\omega_1 = \frac{3}{2}\omega_2$, and a transfer-lattice selection rule organizes the hierarchy (Section 7); the deformation diverges toward the Jordan boundary, with exact leading asymptotics at first order and measured geometric growth beyond (Section 8).

For the perfect-square field theory $S = -\frac{1}{2} \int (\Box\phi + \lambda(\partial\phi)^2)^2$, regulated by a mass splitting: the cubic vertex is exactly the Källén polynomial (Section 9); an even-ghost selection rule protects the positive metric at first order even above the ghost-decay threshold, while the sectorwise branch parity is obstructed there by the real decay (Section 10); at second order the branch-changing scattering $H + L \rightarrow L + L$ obstructs the analytic deformation of the pointed-sector positive metric on an open subset of its shell, with the exact value $401\sqrt{6}/39424$ at a rational kinematic point in fully specified conventions (Section 12); the exact two-field rewriting exhibits the extended theory’s exchange symmetry, its two null Jordan sectors, and the interaction-generated Jordan dynamics (Section 11); and the regulated branch parity, singular on one confluent chain, converges on the doubled space to the linearization of that exchange (Section 13).

Strength of the statements. Throughout we distinguish (a) *analytic metric obstruction* — nonexistence of an order-by-order deformation of the canonical metric in the specified representation and operator class — from (b) *absolute nonexistence of positive metrics*. For the oscillator at 3:1 the stronger statement (b) holds wherever the certified complex pair persists, since a positive metric would make the Hamiltonian similar to a self-adjoint operator; the qualifications (formal perturbation theory, unbounded cubic interaction, truncation) are stated where they apply. For the field theory only (a) is established here. “Generic” is likewise used in three distinct senses, always resolvable from context: *generic oscillator frequencies* (outside the finite resonance-candidate set at the stated order), *generic field obstruction* (nonzero on a nonempty open subset of an allowed scattering shell), and *kinematically open for every mass pair* (existence of the shell, which does not by itself imply nonvanishing of the obstruction at every point of it). The status ledger is: R_1 Jordan scaling — theorem; R_2, R_3 scaling — computational propositions; $R_n \sim \epsilon^{-3n/2}$ — conjecture; the 5:1 obstruction — prediction; a full-field complex spectrum — open; the exact doubled classical involution — theorem; its quantum implementation — open.

Bookkeeping. The deformation order is counted by a formal parameter ζ : $H(\zeta) = H_0 + \zeta H_3 + \zeta^2 H_4$, defining $R_1, R_2, \dots$ and the obstruction orders. The perfect-square coupling λ is a fixed parameter: in the two-field frame the quadratic Jordan structure and the vertex coefficients carry $g = \lambda^2$, while the nonlinear field map itself depends on λ . In the oscillator sections ζ multiplies the cubic vertex and coincides with the coupling.

Verification and reproducibility. Every finite-dimensional, modewise, and distributional identity in this paper is machine-verified in the directory `physics/symplectic-reconstruction` of the repository accompanying the series (release tag `paper5-v1.0`; SymPy 1.14.0, NumPy

2.4.4, Python 3.14). Appendix G maps checks to theorems and classifies each check as exact-symbolic, exact-rational (Sturm/charpoly), or high-precision numerical. Appendices A–F contain readable derivations of the principal results. Operator-domain caveats (unbounded interactions, formal series) are flagged where they occur.

Conventions. Oscillator conventions follow [5, 6]: normalized coordinates $V = (x, y, p, q)$, mode 1 = (x, p) at frequency ω_1 , mode 2 = (y, q) at ω_2 , positive normal form $h_0 = \frac{1}{2}[\omega_1\omega_2 p^2 + (\omega_1/\omega_2)x^2 + (\omega_2/\omega_1)q^2 + \omega_1\omega_2 y^2]$, rapidity $r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}$, $c = \cosh \frac{r}{2} = \omega_1/\sigma$, $s = \sinh \frac{r}{2} = \omega_2/\sigma$, $\sigma = \sqrt{\omega_1^2 - \omega_2^2}$. Operator computations use exact Weyl-symbol Moyal calculus (Appendix A).

2 The deformation complex

Definition 2.1 (Deformation problem and obstruction classes). Let ρ_0 be the canonical free Dyson map, $h_0 = \rho_0 H_0 \rho_0^{-1} = h_0^\dagger$, and $v = \rho_0 V \rho_0^{-1}$ the transported vertex. Seek $\rho_\zeta = \exp[-\frac{1}{2} \sum_n \zeta^n R_n] \rho_0$ with $R_n = R_n^\dagger$ such that $h_\zeta = \rho_\zeta H_\zeta \rho_\zeta^{-1}$ is Hermitian order by order. The first two equations are

$$[h_0, R_1] = v^\dagger - v, \quad [h_0, R_2] = (v_2^\dagger - v_2) + \frac{1}{2}[R_1, v + v^\dagger], \quad (1)$$

where v_2 is the transported $O(\zeta^2)$ vertex when present (derivation: Appendix A). For *discrete* spectrum, $h_0 = \sum_E E P_E$, the equation $[h_0, R] = S$ has a Hermitian solution iff $P_E S P_E = 0$ for every E ; the obstruction class is $\mathfrak{o}_+ = \Pi_{\ker \text{ad}_{h_0}} S = \sum_E P_E S P_E$, and the solution is $R = \sum_{E \neq E'} P_E S P_{E'} / (E - E')$ plus an arbitrary Hermitian element of the commutant. The parallel Krein problem deforms the ghost parity: $[H_0, K_1] = [\kappa_0, V]$, $\{\kappa_0, K_1\} = 0$, with $\mathfrak{o}_K = \Pi_{\ker \text{ad}_{H_0}} [\kappa_0, V]$.

Remark 2.2 (Discrete versus continuous spectrum). The kernel projection above is defined for point spectrum. All field-theoretic computations in this paper are performed at *finite volume* (discrete momenta), where ad_{h_0} has point spectrum and Definition 2.1 applies verbatim; continuum statements are then statements about the finite-volume family, with shell conditions realized by tuning masses to lattice momenta. A distributional (rigged-Fock or spectral-measure) formulation of the continuum obstruction is deferred.

Anti-Hermitian (unitary) generator components are free and do not affect the Hermiticity conditions; commutant additions to R_n are the metric ambiguity. *Gauge independence* of an obstruction class means invariance under both, at all lower orders.

3 The cubic PU oscillator at first order

The real interaction z^3 continues, under the PU rotation $z = iy$, to $V = -iy^3$; the transported vertex is $v = -i(cy + isp)^3$, and $v^\dagger - v = 2ic(c^2y^3 - 3s^2yp^2)$.

Theorem 3.1 (First-order deformation). *For every $\omega_1 > \omega_2 > 0$ the first-order equation is solved by the Hermitian Weyl-ordered cubic*

$$R_1 = \mathcal{W} \left[\frac{2c^3}{\omega_1\omega_2} qy^2 + \frac{4c^3}{3\omega_1^3\omega_2} q^3 - \frac{6cs^2(2\omega_1^2 - \omega_2^2)}{\omega_1\omega_2(4\omega_1^2 - \omega_2^2)} p^2q + \frac{12cs^2\omega_1}{\omega_2(4\omega_1^2 - \omega_2^2)} pxy - \frac{12cs^2\omega_1}{\omega_2^3(4\omega_1^2 - \omega_2^2)} qx^2 \right], \quad (2)$$

with no singular denominator in the physical region; the equivalent Hermitian interaction is $h_1 = \frac{1}{2}(v + v^\dagger) = s(3c^2y^2p - s^2p^3)$. (Verification: substitute (2) into $[h_0, \mathcal{W}[f]] = i\mathcal{W}[\{h_0, f\}]$, exact for quadratic h_0 .)

4 Second order: generic existence

Theorem 4.1 (Generic second-order deformation). *For incommensurate frequencies the second-order obstruction $\Pi_{\ker \text{ad}_{h_0}} \frac{1}{2}[R_1, v + v^\dagger]$ vanishes; the solved R_2 is Hermitian, and the assembled $h_\zeta = e^{-X/2} \star (h_0 + \zeta v) \star e^{X/2}$, $X = \zeta R_1 + \zeta^2 R_2$, is Hermitian through $O(\zeta^2)$, verified end to end in exact Moyal calculus (including the $\frac{1}{8}[[h_0, R_1], R_1]$ term and the Λ^3 contribution). R_2 carries the resonance denominators $(\omega_1 - \omega_2)$ and $(\omega_1 - 3\omega_2)$; no others can vanish for $\omega_1 > \omega_2 > 0$. (Monomial system: Appendix A.)*

5 The exact 3:1 obstruction

Theorem 5.1 (Second-order obstruction at 3:1). *At $\omega_1 = 3\omega_2$ the first-order deformation still exists, but the second-order class is nonzero and exact:*

$$\mathfrak{o}_+^{(2)} = \frac{27\sqrt{3}}{320\omega_2^4} (a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3), \quad (3)$$

the anti-Hermitian on-shell conversion of one mode-1 quantum into three mode-2 quanta. The class is gauge independent under the full relevant first-order freedom: additions to R_1 of Hermitian commutant elements including the quadratic invariants $N_1, N_2, N_1 N_2$ and the resonant quartic operators $a_1 a_2^{\dagger 3} + a_1^\dagger a_2^3$ and $i(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3)$, and anti-Hermitian first-order generators: $\Pi_{\ker \text{ad}_{h_0}}[G, v + v^\dagger] = \Pi_{\ker \text{ad}_{h_0}}[v - v^\dagger, G] = 0$ for all such G . (Second-order kernel additions to R_2 commute with h_0 and cannot affect the second-order class; they enter first at third order.) Derivation: Appendix B.

There is thus no analytic second-order deformation of the canonical positive metric at 3:1, within the Weyl-algebra class, continuously connected to the free metric.

6 Resonant PT breaking

Proposition 6.1 (The antilinear symmetry). *Let $\mathcal{A} = \Pi \circ \Theta$, where Θ is complex conjugation in the (x, y) -Schrödinger representation ($x, y \mapsto x, y$, $p, q \mapsto -p, -q$, $i \mapsto -i$) and Π the total parity ($x, y, p, q \mapsto -x, -y, p, q$). (The symbol Θ is reserved for this conjugation; it is distinct from the Krein parity generators K_n , the effective spectral matrix $K^{(2)}$, and the momentum cutoff $K_{\max}$.) Then $\mathcal{A}$ is antiunitary, $\mathcal{A}^2 = 1$, and*

$$[\mathcal{A}, h_0] = 0, \quad [\mathcal{A}, v] = 0, \quad \text{hence} \quad [\mathcal{A}, h_\zeta] = 0 \quad (\zeta \in \mathbb{R}) : \quad (4)$$

h_0 is even and Θ -real; v is odd under both Π and Θ separately ($\Theta v \Theta = -v$, $\Pi v \Pi = -v$), hence invariant under their composition (machine-verified). “PT breaking” below means the appearance of eigenvalues not fixed by this antilinear symmetry, i.e. complex-conjugate pairs.

Theorem 6.2 (Certified complex pair in the $E = 27\omega_2$ multiplet). *At $\omega_1 = 3\omega_2$ ($\omega_2 = 1$), the second-order effective matrix on the ten-dimensional shell $\{|j, 27 - 3j\rangle\}_{j=0}^9$ is real tridiagonal with antisymmetric off-diagonal $K_{j,j+1} = -\frac{27\sqrt{3}}{640}\sqrt{(j+1)n(n-1)(n-2)}$, $n = 27 - 3j$, and explicit rational diagonal (Appendix C). Because only the products $K_{j,j+1}K_{j+1,j}$ enter, its characteristic polynomial has rational coefficients; an exact Sturm count shows it has exactly eight real roots, hence exactly one complex-conjugate pair, with certified thirty-digit enclosure*

$$\kappa_{\pm} = 2.448807382199383966 \dots \pm 0.224586593808125108 \dots i. \quad (5)$$

Exact diagonalization of the truncated $h_0 + \zeta v$ at $\zeta = 0.02$ (two cutoffs) exhibits the pair with $\text{Im } E = \pm 0.2246 \zeta^2$ to 0.5%. The lowest resonant doublet $\{|1, 0\rangle, |0, 3\rangle\}$ remains real.

Corollary 6.3. *Wherever the complex pair persists in the interacting spectrum, no positive-definite invariant metric exists — analytic in ζ or not. The hierarchy of claims is: (1) the complex pair in the second-order effective resonant-shell matrix is exact (Sturm certificate); (2) its persistence in the interacting truncations is supported by cutoff-stable direct diagonalization; (3) an operator-theoretic statement for the full unbounded cubic Hamiltonian remains open.*

Proposition 6.4 (Broken-PT tongues of the effective multiplet). *Within the second-order effective multiplet, detuning $\delta_r = \omega_1 - 3\omega_2$ adds $j\delta_r$ to the shell diagonal, and the spectrum is complex for $|\delta_r| < \delta_c(\zeta) = 38.4151 \dots \times \zeta^2 + o(\zeta^2)$. That an order- n resonance produces an $O(\zeta^n)$ tongue is the corresponding general perturbative expectation, not a proved proposition.*

7 Third order and the selection rules

Theorem 7.1 (Transfer-lattice selection rule). *Grade monomials by energy transfer, $[h_0, M] = [(\Delta n_1)\omega_1 + (\Delta n_2)\omega_2]M$. Order- n objects have transfers in the n -fold sumset of the vertex transfer set, total degree $\leq n + 2$, and degree parity $\equiv n \pmod{2}$ (proof: Appendix D). The order- n obstruction is supported on the finite candidate sets: $\{2:1\}$ at order 1, $\{3:1\}$ at order 2, $\{3:2, 2:1, 4:1\}$ at order 3, with 5:1 first reachable at order 4. The lattice gives candidates; coefficients decide.*

Theorem 7.2 (Third order). *At generic frequencies the third-order obstruction vanishes and R_3 is Hermitian with odd total degrees ≤ 5 . At $\omega_1 = \frac{3}{2}\omega_2$ the class is nonzero; at $(\omega_1, \omega_2) = (3, 2)$ its exact value is*

$$\mathfrak{o}_+^{(3)} = -\frac{117\sqrt{30}}{1120} i (a_1^2 a_2^{\dagger 3} + a_1^{\dagger 2} a_2^3), \quad (6)$$

and the coefficient scales exactly as $\omega_2^{-13/2}$ (verified by the ratio $2^{-13/2}$ between $(\omega_1, \omega_2) = (6, 4)$ and $(3, 2)$), so the general form carries the dimensional factor $\omega_2^{-13/2}$. The class is gauge independent in the sense of Theorem 5.1. The candidates 2:1 (orders 1 and 3) and 4:1 (order 3) have vanishing coefficients.

Conjecture 7.3 (Resonance hierarchy). *The coprime ratio $\omega_1/\omega_2 = p:q$ first obstructs at order $p+q-2$ when the mode-2 transfer p is odd. (Verified: 3:1 at order 2, 3:2 at order 3. Predicted: 5:1 at order 4, 5:3 at order 6. The mechanism excluding even mode-2 transfers is open.) The resonance set is an expanding hierarchy, not (so far) a dense set.*

8 The Jordan divergence hierarchy

Theorem 8.1 (Exact first-order Jordan asymptotics). *With $\omega_{1,2} = \omega \pm \epsilon$, the explicit R_1 of (2) has the exact leading Laurent behaviour*

$$R_1 = \epsilon^{-3/2} \mathcal{W} \left[\frac{xyz}{2\sqrt{\omega}} + \frac{y^2q}{4\sqrt{\omega}} - \frac{p^2q}{4\sqrt{\omega}} - \frac{x^2q}{2\omega^{5/2}} + \frac{q^3}{6\omega^{5/2}} \right] + O(\epsilon^{-1/2}), \quad (7)$$

so $R_1 = \Theta(\epsilon^{-3/2})$, while the free generator diverges only logarithmically.

Computational Proposition 8.2 (Higher orders). *The solved R_2 and R_3 scale as ϵ^{-3} and $\epsilon^{-9/2}$ (measured decade exponents -3.00 and -4.49 over $\epsilon = 10^{-2} \rightarrow 10^{-3}$), with no small-denominator enhancement from $\omega_1 - \omega_2$ at second order. These are high-precision numerical statements at fixed orders, not proved big- O estimates.*

Conjecture 8.3. $R_n = \Theta(\epsilon^{-3n/2})$, so the deformation's radius of convergence in ζ collapses as $\zeta_c \sim \epsilon^{3/2}$ toward the Jordan boundary.

9 The perfect-square vertex

The interacting theory is $S = -\frac{1}{2} \int (\Box\phi + \lambda(\partial\phi)^2)^2$ [3], with cubic vertex $V_3 \propto \Box\phi(\partial\phi)^2$ and quartic $V_4 \propto \frac{1}{2}(\partial\phi)^4$.

Proposition 9.1 (Källén form). *On $p_1 + p_2 + p_3 = 0$ the symmetrized cubic vertex is exactly*

$$\mathcal{V}_3 = p_1^2(p_2 \cdot p_3) + p_2^2(p_1 \cdot p_3) + p_3^2(p_1 \cdot p_2) = \frac{1}{2} \lambda_K(p_1^2, p_2^2, p_3^2). \quad (8)$$

The polynomial vertex factor therefore has a second-order zero at the massless confluent point: $\lambda_K(0, 0, 0) = 0$ and $\nabla \lambda_K(0, 0, 0) = 0$, while the Hessian is nonzero. On the regulated decay channel, $\lambda_K(m_1^2, m_2^2, m_2^2) = m_1^2(m_1 - 2m_2)(m_1 + 2m_2)$: the vertex vanishes exactly at threshold and turns on continuously above it. (A full external-state selection rule for generalized Jordan legs would additionally require the confluent limit of leg normalizations, shell measures, and mode factors; we do not claim it here.)

10 First order in the regulated field theory

Regulate by $P_\delta = (\Box + m_1^2)(\Box + m_2^2)$, $\delta = m_1^2 - m_2^2 > 0$, at finite volume (Remark 2.2). In the mode expansion every leg is on its branch shell, so on the kernel the vertex coefficient is $\frac{1}{2}\lambda_K$ of the branch masses. The transported field mode, derived from the free Dyson transport and verified against the spectral Wightman function of [7] (Appendix E), is

$$\phi'(k) = \frac{i}{\sqrt{2\delta}} \left[\frac{a_2 + a_2^\dagger}{\sqrt{\omega_2}} + \frac{a_1 - a_1^\dagger}{\sqrt{\omega_1}} \right], \quad \sigma = \sqrt{\delta} \text{ (} k\text{-independent)}. \quad (9)$$

Theorem 10.1 (First-order protection; sectorwise parity obstruction). *(i) Even-ghost selection rule: the transported reality factors make $v^\dagger - v$ vanish on every odd-ghost-count monomial (Appendix E). Even-ghost shells are kinematically closed for all $m_1 > m_2 > 0$. Hence $\mathfrak{o}_+^{(1)} = 0$ identically — below and above the decay threshold. (ii) The sectorwise branch parity $\kappa_0 = (-1)^{N_{\text{ghost}}}$ has $\mathfrak{o}_K^{(1)} = 0$ below threshold and, above it, a nonzero class on the open $1 \rightarrow 22$ shell, equal there to $\lambda_K(m_1^2, m_2^2, m_2^2)$ times the leg normalizations.*

Computational Proposition 10.2 (Split-basis scaling). *At massive confluence both R_1 and K_1 diverge as $\delta^{-3/2}$; on massless Jordan paths $m_2^2 = \alpha\delta$ the positive R_1 scales as $\delta^{-1/2}$ with path-independent power, while the Krein K_1 is $\delta^{-1/2}$ generically and $\delta^{-3/2}$ on the exceptional path $\alpha = 2/7$. These are statements about coefficients in the split mode basis, which itself degenerates at $\delta = 0$; a basis-independent formulation on confluent states is deferred.*

11 The exact two-field rewriting

Theorem 11.1 (Two-field form and exchange symmetry). *Introducing the auxiliary field ψ and the change of variables $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$ (unit pointwise Jacobian),*

$$\mathcal{L} = -\partial_\mu U \partial^\mu V + \frac{\lambda^2}{2} U^2 V^2 = -\frac{1}{2}(\partial X)^2 + \frac{1}{2}(\partial Y)^2 + \frac{\lambda^2}{8}(X^2 - Y^2)^2, \quad X, Y = \frac{U \pm V}{\sqrt{2}}. \quad (10)$$

The exchange $U \leftrightarrow V$ is an exact symmetry of the extended two-field theory. In the original variables it is the transition map between the charts $\phi_A = \frac{1}{\lambda} \log U$ and $\phi_B = \frac{1}{\lambda} \log V$,

$$(\phi, \psi) \mapsto \left(\frac{1}{\lambda} \log \frac{\psi}{\lambda} - \phi, \psi \right), \quad (11)$$

defined on the chart overlap ($U \neq 0 \neq V$), singular at the perturbative vacuum $\psi = 0$, and requiring a branch choice in the complexified theory. The original (ϕ, ψ) chart covers only $U \neq 0$ and does not contain the mirror background: the exchange is a global symmetry of the extension, not an everywhere-defined involution of the original field space, and it is invisible at any finite perturbative order around the original vacuum.

Proposition 11.2 (Null sectors and interaction-generated Jordan dynamics). *The constant zero-action stationary backgrounds form the two null branches $UV = 0$ (whether they are vacua in a given completion is a separate question, since the kinetic form is indefinite). The perturbative background $(U, V) = (1, 0)$ is a nonzero null configuration; the exchange maps it to $(0, 1)$. Expanding about $(1, 0)$: $\mathcal{L}^{(2)} = -\partial u \partial v + \frac{g}{2} v^2$, $\mathcal{L}^{(3)} = g u v^2$, $\mathcal{L}^{(4)} = \frac{g}{2} u^2 v^2$ ($g = \lambda^2$), with equations $\square v = 0$, $\square u = g v$: the exact $\square^2$ Jordan pair, the off-diagonal supplied by the interaction coupling; about $(0, 1)$ the mirror expansion has the oppositely oriented chain. A chain-preserving involution commuting with a single Jordan block is $\pm I$ (companion lemma), so exchange between oppositely oriented sectors is the only possible realization of a parity-like symmetry. In this frame the interactions are pure potentials: $H_{\text{int}} = -L_{\text{int}}$ exactly, with no Ostrogradsky–Legendre corrections.*

12 The sectorwise second-order obstruction

Regulate the sector- A theory by $\mathcal{L}_0 = -\partial u \partial v - \mu^2 u v + \frac{g}{2} v^2 + \frac{\epsilon}{2} u^2$, with masses $m_\pm^2 = \mu^2 \pm \sqrt{\epsilon g}$ and branch eigenvectors $v = \rho_b u$, $\rho_\pm = \mp \delta / (2g)$, $\delta = m_+^2 - m_-^2$ (the regulator necessarily breaks the exchange symmetry). All computations below set $g = 1$. The positive-frame quantization and the complete normalization conventions — finite volume with unit-normalized discrete modes $[a, a^\dagger] = 1$, unit-norm external states, lattice momenta, $m_L = 4$ — are collected in Appendix E. The second-order on-shell source element is computed from the exact matrix-element form

$$\langle \text{out} | \mathcal{S}_+^{(2)} | \text{in} \rangle = \langle v_2^\dagger - v_2 \rangle + \sum_n \frac{\langle \text{out} | v_1^\dagger | n \rangle \langle n | v_1^\dagger | \text{in} \rangle - \langle \text{out} | v_1 | n \rangle \langle n | v_1 | \text{in} \rangle}{E - E_n}, \quad (12)$$

derived in Appendix E from $(v^\dagger - v)(v + v^\dagger) + (v + v^\dagger)(v^\dagger - v) = 2(v^\dagger v^\dagger - vv)$.

Theorem 12.1 (Sectorwise obstruction on an open shell subset). *Consider the branch-changing scattering $H + L \rightarrow L + L$, kinematically open at suitable momenta for every $m_H > m_L$. At the rational center-of-mass point $m_H = \frac{3}{2}m_L$ (both incoming particles at rest; outgoing momenta $\pm \frac{3}{4}m_L$; in lattice units $m_L = 4$, $m_H = 6$, momenta ± 3 , all shell conditions exact), the source element (12) has the exact value*

$$\mathcal{M}_{\text{obs}} = \frac{401\sqrt{6}}{39424 g^2} \xrightarrow{g=1} \frac{401\sqrt{6}}{39424} \quad (13)$$

(the $1/g^2$ dependence at fixed masses is verified exactly: each vertex carries $g\rho^2 \propto \delta^2/g$), with the quartic contact piece exactly zero. The computation is truncation-complete (Lemma 12.2) and the element is analytic in the shell data on the nonsingular part of the shell; hence the second-order obstruction is nonzero on a nonempty open subset of the shell containing this point. Independent tuned kinematic points give 0.5233, 0.3109, 0.3970 (numerical), consistent with nonvanishing on a large region; a characterization of the zero set is left open, and no claim is made for every mass pair.

Lemma 12.2 (Truncation completeness). *For external momenta $\{k_a, k_b\} \rightarrow \{k_c, k_d\}$ pairwise distinct, every intermediate Fock state contributing to (12) has all momenta in the finite set generated by the externals and their pairwise sums and differences ($s = k_a + k_b$, $t = k_a - k_c$, $u = k_a - k_d$): in each contributing chain the surviving spectator particles must coincide with external particles, which fixes every internal momentum; free-momentum loops arise only in forward/disconnected kinematics, excluded here. (Proof: Appendix E; corroborated by exact cutoff independence between $K_{\text{max}} = 4$ and 6.)*

Remark 12.3 (Oscillator versus field theory).

PU oscillator	field theory
discrete energy spectrum	continuous momentum spectrum
isolated rational resonance	generic scattering shell
3:1 first appears at order 2	$H + L \rightarrow L + L$ appears at order 2
finite resonant multiplet	continuum of degenerate in/out states

The continuum turns the discrete resonance obstruction into a scattering-shell obstruction. The exact exchange symmetry does not cure it: it maps the obstructed sector- A construction to the mirror sector- B construction.

13 The confluent parity theorem

Theorem 13.1 (Confluent limit of the regulated parity). *Let $P_\delta b_\pm = \pm b_\pm$ be the regulated branch parity and $c = (b_+ + b_-)/2$, $d = (b_+ - b_-)/\delta$ the confluent Jordan basis, so $P_\delta c = \frac{\delta}{2}d$ and $P_\delta d = \frac{2}{\delta}c$; in the standard column convention on the ordered basis (c, d) ,*

$$P_\delta = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}, \quad P_\delta^2 = 1, \quad (14)$$

with no bounded limit as $\delta \rightarrow 0$: the regulated parity cannot converge as a chain-preserving parity inside one Jordan sector. On the doubled space with the oppositely oriented confluent

identification in the mirror sector ($c_B = (b_+ - b_-)/2$, $d_B = (b_+ + b_-)/\delta$), the cross parity $b_{\pm}^A \leftrightarrow \pm b_{\pm}^B$ equals, exactly and δ -independently, the sector exchange $(c_A, d_A) \leftrightarrow (c_B, d_B)$. The regulated branch parity thus converges, after doubling by the oppositely oriented Jordan sector, to the linearization of the exact sector-exchange involution around the two mirror backgrounds. (Identifying the full nonlinear quantum operator requires a separate construction, not supplied here.)

14 Discussion

The mechanism. Away from resonances the positive pseudo-Hermitian metric deforms locally, with explicit generators. The failures are structural: on-shell branch-changing processes produce gauge-independent cohomological obstructions — isolated rational loci for the oscillator, generic scattering shells in the field theory; at the oscillator resonances the obstruction is certified spectral; and independent of resonances, the deformation loses uniformity toward Jordan degeneration. The sectorwise branch parity is broken by real ghost decay above threshold; what survives at the massless perfect-square boundary is the exact sector-exchanging involution of the extended two-field theory, whose linearization is the bounded confluent limit of the regulated parity on the doubled space.

Symmetry enhancement at the boundary. The massless perfect-square point is not the split theory with parameters set to zero: the Källén vertex degenerates there, the doubled $O(1,1)$ structure emerges, and the Jordan dynamics is generated by the interaction about its own null backgrounds — a boundary-emergent structure mirroring the Weyl enhancement at the conformal boundary of the gravity companion [8].

Open questions. Whether the two pointed sectors admit parity vacua ($|0_A\rangle \pm |0_B\rangle$)/ $\sqrt{2}$ or are superselected requires finite-volume overlap and zero-mode analysis; the quantum implementation of the exchange on a completion, and any probability interpretation, remain open. The hierarchy conjecture predicts a 5:1 obstruction at fourth order. The gravitational extension — the cubic Einstein–Weyl vertices against the covariant real forms of [8] — is the natural next target: the machinery here makes branch-changing gravitational scattering the decisive calculation. No CPT claim is made: the exchange is a linear sector map, not a constructed antiunitary CPT operator.

References

- [1] C. M. Bender and P. D. Mannheim, *No-ghost theorem for the fourth-order derivative Pais–Uhlenbeck oscillator model*, Phys. Rev. Lett. **100** (2008) 110402.
- [2] C. M. Bender and P. D. Mannheim, *Exactly solvable PT-symmetric Hamiltonian having no Hermitian counterpart*, Phys. Rev. D **78** (2008) 025022, arXiv:0804.4190.
- [3] S. Bateman and N. Turok, *Escape from Ostrogradsky via hidden ghost parity*, arXiv:2607.00096.
- [4] A. Mostafazadeh, *Metric operator in pseudo-Hermitian quantum mechanics and the imaginary cubic potential*, J. Phys. A **39** (2006) 10171, arXiv:quant-ph/0508195.

- [5] *Canonical positive symplectic diagonalization of the Pais–Uhlenbeck oscillator*, companion paper (Paper I) and verification repository, 2026.
- [6] *The Pais–Uhlenbeck metric as a minimum-distortion principle, and the representation problem for the fourth-order field*, companion paper (Paper II), 2026.
- [7] *The universal vacuum of the fourth-order scalar field: metric orbits, Fock sectors, and the Krein boundary*, companion paper (Paper III), 2026.
- [8] *Gauge reduction and the completion problem in fourth-order gravity: PU pairing, covariant real forms, and the conformal Jordan boundary*, companion paper (Paper IV), 2026.
- [9] T. Kato, *Perturbation Theory for Linear Operators*, Springer, 1966.
- [10] W. D. Heiss, *The physics of exceptional points*, J. Phys. A **45** (2012) 444016.

A The Weyl–Moyal recursion

Operators are represented by Weyl symbols on the two canonical pairs (x, p) , (y, q) ; the star product is $f \star g = \sum_n \frac{1}{n!} (i/2)^n \Lambda^n(f, g)$ with $\Lambda = \overleftarrow{\partial}_x \overrightarrow{\partial}_p - \overleftarrow{\partial}_p \overrightarrow{\partial}_x + \overleftarrow{\partial}_y \overrightarrow{\partial}_q - \overleftarrow{\partial}_q \overrightarrow{\partial}_y$, a finite sum on polynomials. For quadratic h_0 , $[h_0, \mathcal{W}[f]] = i\mathcal{W}[\{h_0, f\}]$ exactly; for cubic–cubic commutators the Λ^3 (constant) term contributes. Expanding $h_\zeta = e^{-X/2} \star (h_0 + \zeta v) \star e^{X/2}$, $X = \zeta R_1 + \zeta^2 R_2 + \zeta^3 R_3$, with $e^{-X/2} A e^{X/2} = A + \frac{1}{2}[A, X] + \frac{1}{8}[[A, X], X] + \frac{1}{48}[[[A, X], X], X] + \dots$, the order- ζ^2 and ζ^3 Hermiticity conditions are

$$[h_0, R_2] = (v_2^\dagger - v_2) + \frac{1}{2}[R_1, v + v^\dagger], \quad (15)$$

$$[h_0, R_3] = -(T_3 - T_3^\dagger), \quad (16)$$

$$T_3 = \frac{1}{8}([h_0, R_1], R_2) + \frac{1}{8}([h_0, R_2], R_1) + \frac{1}{48}([h_0, R_1], R_1), R_1 + \frac{1}{2}[v, R_2] + \frac{1}{8}[[v, R_1], R_1],$$

using $[h_0, R_1] = v^\dagger - v$ and the fact that $[v - v^\dagger, R_1]$ is Hermitian. Equations $[h_0, R] = S$ are solved monomial by monomial in the mode variables $a_1^m a_1^{\dagger n} a_2^r a_2^{\dagger s}$, where $[h_0, M] = \Omega_M M$ with $\Omega_M = (n - m)\omega_1 + (s - r)\omega_2$: $R_M = S_M/\Omega_M$ off the kernel; the kernel component is the obstruction.

B The 3:1 projection and gauge independence

At $\omega_1 = 3\omega_2$ the second-order source $S_2 = \frac{1}{2}[R_1, v + v^\dagger]$ is a quartic polynomial; its kernel monomials must satisfy $(n - m)3\omega_2 + (s - r)\omega_2 = 0$ with degree ≤ 4 , i.e. $(m - n, r - s) = \pm(1, -3)$, realized only by $a_1 a_2^{\dagger 3}$ and $a_1^\dagger a_2^3$. Direct evaluation of the two coefficients gives $\mathbf{o}_+^{(2)} = \frac{27\sqrt{3}}{320\omega_2^2}(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3)$. Gauge independence: a change $R_1 \rightarrow R_1 + G$ with $[G, h_0] = 0$ shifts S_2 by $\frac{1}{2}[G, v + v^\dagger]$; transfer bookkeeping shows that for every Hermitian kernel generator (N_1, N_2) polynomials and the resonant quartics $a_1 a_2^{\dagger 3} + a_1^\dagger a_2^3, i(a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3)$ the shifted source has no kernel component: the transfers of $v + v^\dagger$, shifted by $\pm(1, -3)$ or $(0, 0)$, never land on $\pm(1, -3)$; machine verification confirms $\Pi_{\ker \text{ad}_{h_0}}[G, v + v^\dagger] = 0$ term by term. Anti-Hermitian generators A_1 enter through $\frac{1}{2}[v - v^\dagger, A_1]$, and the same bookkeeping gives $\Pi_{\ker \text{ad}_{h_0}}[v - v^\dagger, A_1] = 0$. Kernel additions to R_2 drop out of the second-order equation since $[h_0, R_2^{\ker}] = 0$.

C The effective shell matrix

Standard second-order degenerate perturbation theory on the shell $\mathcal{E}_{27} = \text{span}\{|j, 27 - 3j\rangle\}$ gives $K_{ab}^{(2)} = \sum_{m \notin \mathcal{E}} v_{am} v_{mb} / (27 - E_m)$ (the first-order block vanishes: cubic v cannot realize the transfer $\pm(1, -3)$). Evaluating the sums yields the rational diagonal $D(n_1, n_2) = (633n_1^2 - 1836n_1n_2 - 285n_1 + 3969n_2^2 + 3051n_2 + 818)/26880$ and the antisymmetric off-diagonal quoted in Theorem 6.2. For a tridiagonal matrix only the products $K_{j,j+1}K_{j+1,j} \in \mathbb{Q}$ enter the characteristic polynomial, computed exactly by the three-term recurrence $p_j = (D_j - \kappa)p_{j-1} - K_{j-1,j}K_{j,j-1}p_{j-2}$; the Sturm count of the resulting degree-10 rational polynomial is exactly 8.

D The transfer lattice

Transfer additivity: the grading by ad_{h_0} satisfies $[h_0, AB] = [h_0, A]B + A[h_0, B]$, so transfers add under operator products, hence under star products and commutators. Degree bound: each commutator of polynomial factors lowers total degree by at least two (Λ^1), so order- n objects, built from n cubic factors and $n - 1$ commutators (with $[h_0, \cdot]$ degree-preserving), have degree $\leq 3n - 2(n - 1) = n + 2$; Moyal corrections lower degree by even amounts, fixing the parity $\equiv n \pmod{2}$. A kernel monomial with transfer (d_1, d_2) , $d_1 d_2 < 0$, requires degree $\geq |d_1| + |d_2|$ with matching per-mode parities, giving the candidate sets of Theorem 7.1.

E Field conventions, the source formula, and truncation completeness

Conventions. Finite volume; discrete momenta k (lattice units with $m_L = 4$ in Theorem 12.1); modes (b, k) with $[a_{b,k}, a_{b',k'}^\dagger] = \delta_{bb'} \delta_{kk'}$; unit-norm external states; $g = 1$; deformation parameter ζ multiplying H_3 , ζ^2 multiplying H_4 . Positive-frame legs from the branch eigenvectors: u -legs $(a_+ + a_+^\dagger)/\sqrt{2\omega_+ \delta}$ (healthy) and $(a_- - a_-^\dagger)/\sqrt{2\omega_- \delta}$ (ghost, quarter-turn); v -legs $\rho_\pm$ times u -legs, $\rho_\pm = \mp \delta/(2g)$. Interactions $H_3 = -g \int uv^2$, $H_4 = -\frac{g}{2} \int u^2 v^2$ are pure potentials. The transported fourth-order field of Section 10 is fixed by matching the free two-point function $[e^{-i\omega_1 t}/2\omega_1 - e^{-i\omega_2 t}/2\omega_2]/\delta$ of [7]; the even-ghost rule follows because a monomial with g ghost legs, c_g of them creators, has v -coefficient $\propto i^{\# \text{legs}} (-1)^{c_g}$ and $v^\dagger$ -coefficient $\propto$ the conjugate with $c_g \rightarrow g - c_g$, whose difference vanishes for odd g .

Source formula. With $E_{\text{in}} = E_{\text{out}} = E$ and R_1 solved off shell, $\langle \text{out} | [R_1, w] | \text{in} \rangle = \sum_n [(v^\dagger - v)_{\text{out},n} w_{n,\text{in}} + w_{\text{out},n} (v^\dagger - v)_{n,\text{in}}] / (E - E_n)$ with $w = v + v^\dagger$; the algebraic identity $(v^\dagger - v)w + w(v^\dagger - v) = 2(v^\dagger v^\dagger - vv)$ reduces this to (12).

Truncation completeness (proof of Lemma 12.2). Second-order chains pass through 1-, 3-, or 5-particle intermediates. 1-particle: momentum $k_a + k_b$ fixed. 3-particle: the first vertex annihilates one external and creates a pair $(q, k_x - q)$; for the second vertex to reach the 2-particle out-state, the surviving created particle must be an out-particle, forcing $q \in \{k_c, k_d, k_x - k_c, k_x - k_d\}$; annihilating both created particles instead returns the in-momenta, giving zero overlap with a distinct out-state. 5-particle: the created triple (p, q, r) , $p + q + r = 0$, must contain both out-particles, fixing the third momentum. Hence all internal momenta lie in the s, t, u -generated set, and enlarging the momentum cutoff cannot change the element — as the exact agreement at $K_{\text{max}} = 4, 6$ confirms.

F The doubled confluent parity

With change-of-basis matrix C whose columns are (c, d) in branch coordinates, $P_\delta^{(c,d)} = C^{-1} \text{diag}(1, -1)C = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}$. For the doubled space take $C_A = C$ and the oppositely oriented C_B (columns $c_B = (b_+ - b_-)/2$, $d_B = (b_+ + b_-)/\delta$); the cross map $\kappa(b_\pm^A) = \pm b_\pm^B$ has matrix $T^{-1} \kappa_{\text{branch}} T$ with $T = \text{diag}(C_A, C_B)$, which evaluates to the δ -independent exchange $(c_A, d_A) \leftrightarrow (c_B, d_B)$, squaring to the identity.

G Verification checks and their classification

Check	Supports	Type
ID1–ID5	Theorem 3.1	exact symbolic
ID6–ID8	Theorem 4.1	exact symbolic
ID9a–b	Theorem 5.1	exact symbolic
ID10, O3e	Comp. Prop. 8.2	high-precision numeric
SR1–SR3	Theorem 7.1	exact combinatorial
O3a–O3d	Theorem 7.2	exact symbolic
PT1–PT2	Theorem 6.2 (structure)	exact symbolic
(new) Sturm count	Theorem 6.2 (pair)	exact rational
PT3	Theorem 6.2 (truncated op.)	numeric
PT4	broken-PT tongues	numeric (effective matrix)
PS1–PS2	Proposition 9.1	exact symbolic
PS-A–PS-G	Theorem 10.1, Comp. Prop.	exact symbolic + numeric
PS-H	Jordan-chain lemma	exact symbolic
TF1–TF7	Theorem 11.1, Prop. 11.2	exact symbolic
SO1–SO7	Theorem 12.1 (numerics)	exact + numeric
HX1	Theorem 12.1 (exact value)	exact symbolic
HX2–HX3	Theorem 13.1	exact symbolic
(new) PT proposition	Proposition 6.1	exact symbolic
(new) $\omega_2^{-13/2}$ ratio	Theorem 7.2	exact symbolic
(new) R_1 Laurent	Theorem 8.1	exact symbolic

Scripts, in symbolic/, each printing ALL PASS:

```
verify_interaction_deformation.py  verify_interaction_order3.py
verify_pt_breaking.py  verify_perfect_square.py  verify_two_field.py
verify_sector_obstruction.py  verify_hardening.py
```